Year 12 Maths - Vectors

Similar questions - Magnitude, direction and components

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) The vector $\mathbf{a}$ has magnitude 10 and direction 150° measured anticlockwise from the positive x-axis. Write $\mathbf{a}$ in the form $p\mathbf{i}+q\mathbf{j}$, giving exact values of p and q. **[3]**

(b) The vector $\mathbf{b}=3\mathbf{i}-4\mathbf{j}$. Find $|\mathbf{b}|$ and the angle between $\mathbf{b}$ and the positive x-axis, to 1 decimal place. **[3]**

(c) Find $|\mathbf{a}+\mathbf{b}|$, to 3 significant figures. **[2]**

2 The vector $\mathbf{p}=k\mathbf{i}+(2k-1)\mathbf{j}$, where k is a constant, has magnitude $\sqrt{10}$.

(a) Show that $5k^2-4k-9=0$. **[2]**

(b) Find the two possible values of k and the corresponding vectors $\mathbf{p}$. **[3]**

(c) For the negative value of k, find a unit vector in the direction of $\mathbf{p}$. **[2]**

Question 1

(a) $\mathbf{a} = 10\cos 150^\circ\,\mathbf{i} + 10\sin 150^\circ\,\mathbf{j} = -5\sqrt{3}\,\mathbf{i} + 5\mathbf{j}$

(b) $|\mathbf{b}| = \sqrt{9+16} = 5$. Angle below the positive x-axis: $\tan^{-1}\frac{4}{3} = 53.1^\circ$ (i.e. 306.9° anticlockwise)

(c) $\mathbf{a}+\mathbf{b} = (3-5\sqrt{3})\mathbf{i}+\mathbf{j}$. $|\mathbf{a}+\mathbf{b}|^2 = (3-5\sqrt{3})^2+1 = 85-30\sqrt{3}$, so $|\mathbf{a}+\mathbf{b}| = 5.75$

Question 2

(a) $k^2+(2k-1)^2 = 10$, so $5k^2-4k+1 = 10$ and $5k^2-4k-9 = 0$

(b) $(5k-9)(k+1) = 0$: $k = \frac{9}{5}$ giving $\mathbf{p} = \frac{9}{5}\mathbf{i} + \frac{13}{5}\mathbf{j}$, or $k = -1$ giving $\mathbf{p} = -\mathbf{i} - 3\mathbf{j}$

(c) $\dfrac{1}{\sqrt{10}}(-\mathbf{i} - 3\mathbf{j})$